

CHANGING CLIMATES

The features of our seasons – for example, hot, dry summers and cold, icy winters – make up the climate of each region. Earth's climate has always changed at different rates, places, and times, and this variation in conditions has considerable and continuing impact upon the evolution of life.

WHAT IS CLIMATE?

Climate is a region's average weather over a long period of time, produced by all the atmospheric conditions such as temperature, rainfall, wind strength, and pressure. The climate of any given place is also partly controlled by other factors such as human input, height above sea level, local topography, proximity to seas and oceans – with

their prevailing winds and water currents – and, most importantly, its latitudinal position between the equator and pole. Latitude controls the amount of solar radiation received

< CHANGING TREE LINE

As the air temperature declines with increasing altitude, plant life changes. Broadleaved trees are replaced by conifers, then shrubs.



in different regions of the world. For instance, there is a major difference between climates of polar regions, which receive the least light and heat, and of tropical regions around the equator, which receive the most.

CHANGING CONDITIONS

Global climates are generally classified according to the average temperature and the amount of rainfall each area receives, and their combined effect on plant growth. For instance, equatorial regions at present are hot and wet because oceans dominate there, whereas deserts are dry and polar regions are cold. However, this has not always been the case. Factors controlling weather conditions over geological time have affected the climate of the planet, from ice ages through to global warming.



ARCTIC FOX
IN SUMMER



ARCTIC FOX
IN WINTER

^ SEASONAL ADAPTATION

Annual climate change can bring acutely different living conditions from one season to the next. Animals and plants have various means of adapting to these changes; for example, the Arctic fox grows a thick coat in winter, which it sheds in summer.

LIFE IN THE DESERT

Plants such as cacti have adapted to life in semi-arid environments by slowing growth, reducing the leaves to spines (to stop water loss from transpiration), and evolving special tissues that retain water.

